Sectional, assembly & exploded views

GRAPHICS WITH MR KELLY

Higher · Drawing & spatial awareness · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ read and draw a sectional view with a labelled cutting plane
- ☐ apply hatching correctly, leaving webs, shafts, nuts and bolts unhatched
- ☐ describe good and poor practice in an exploded view
- ☐ recognise full, half, part, revolved, removed and stepped sections and cut-aways

I DO — Worked example

Read how a full-mark answer is built

An exploded pictorial of a desk lamp is shown.

1 Describe two features of good practice in an exploded view. (2 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Parts are spaced so none overlap
- ✓ All parts can be seen

WE DO — Guided practice

Use the starters and word bank to help you

A production drawing of a crank arm includes several types of sectional view.

2 State the name of two types of section other than a full section, and describe when each is used. (2 marks)

Sentence starters

Part section: …

Revolved section: …

Word bank

Stepped section · cutting · Part section · cross-section · Revolved section · feature · rotated · internal

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which is a feature of a good exploded view?
☐ Parts are aligned along assembly lines ☐ Parts overlap to save space on the page ☐ Parts are shown in any order that fits ☐ Hidden detail is shown on every part

b) Which of these is NOT hatched when cut lengthways?
☐ A casing ☐ A base plate ☐ A bolt ☐ A housing

c) A cross-section rotated 90° on the view itself is a...
☐ Part section ☐ Full section ☐ Revolved section ☐ Cut-away

d) Hatching lines are normally drawn at...
☐ 30° ☐ 90° ☐ 60° ☐ 45°

4 Key terms: write the letter of the matching description next to each term.

1. **Hatching**	____	**A** Correct order, aligned, evenly spaced, no overlaps, all parts visible.
2. **Which parts are not hatched?**	____	**B** Webs, ribs, shafts, nuts, bolts, screws, washers, pins and threads cut lengthways.
3. **SECTION A–A**	____	**C** Thin 45° lines on cut surfaces of a sectional view.
4. **Good exploded view**	____	**D** Cross-section rotated 90° and drawn on the view.
5. **Revolved section**	____	**E** The label for the view taken at cutting plane A–A.

A sectional assembly of a pulley shows a casing, a shaft, a nut and bolt, and a strengthening web.

5 Explain which parts should be hatched and which should not. (3 marks)

6 Explain what is meant by ‘Stepped section’. (2 marks)

7 Explain what is meant by ‘Part section’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?